Student Performance Prediction

Using Machine Learning

This project aims to predict a student's academic performance at an early stage using machine learning techniques. The main idea is to help students estimate their expected final results before exams, allowing them to organize their study plans more effectively.

Additionally, the project helps educators identify students who may be at academic risk and require early intervention.

Slide 2

Dataset Overview

The dataset used in this project was collected from Kaggle and is known as the Student Performance Dataset.

6608
Students (Rows)

20
Features (Columns)

7
Numerical Features

13
Categorical Features

Dataset Characteristics

The dataset contains academic, social, and personal attributes related to students. It is structured and suitable for supervised machine learning tasks.

Slide 3

Data Preprocessing

Essential steps applied before training the machine learning models

Step 1

Missing Values Check

The dataset was checked for missing values to ensure data quality

Step 2

Categorical Conversion

Categorical features were converted into numerical values

Step 3

One-Hot Encoding

Applied for nominal categorical features

Step 4

Label Encoding

Applied where numerical labels were appropriate

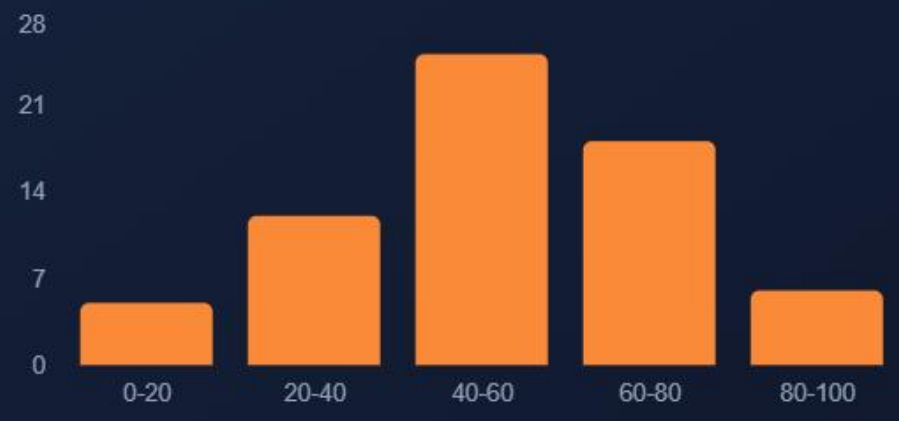
✓ These steps helped improve model accuracy and stability

Data Visualization

Understanding the dataset through visual analysis

Categorical Feature Distribution

Numerical Feature Histogram



Key Insights

Visualization helped identify trends and relationships affecting student performance. This step supported better feature understanding before modeling.

Slide 5

Machine Learning Models

Four algorithms implemented and trained on the preprocessed dataset

Linear Regression

Simple yet powerful baseline model for linear relationships

Ridge Regression

L2 regularized linear model preventing overfitting

Random Forest Regressor

Ensemble of decision trees for robust predictions

XGBoost Regressor

Gradient boosting for high-performance prediction

All models were trained on the same preprocessed dataset to ensure fair comparison

Slide 6

Evaluation Metrics

Metrics used to compare accuracy and prediction reliability

MSE
Mean Squared Error

Measures average squared difference between predicted and actual values

$$MSE = (1/n)\ \Sigma(y_i - \hat{y}_i)^2$$

RMSE
Root Mean Squared Error

Square root of MSE, in the same units as the target variable

$$RMSE = \sqrt{MSE}$$

R^2 Score
Coefficient of Determination

Proportion of variance explained by the model (0-100%)

$$R^2 = 1 - (SS_{res} / SS_{tot})$$

These metrics were used to compare accuracy and prediction reliability across all models

Model Results

Linear Regression

RMSE	≈0.54
R^2	97%

★ Best Performance

Ridge Regression

RMSE	≈0.54
R^2	97%

★ Best Performance

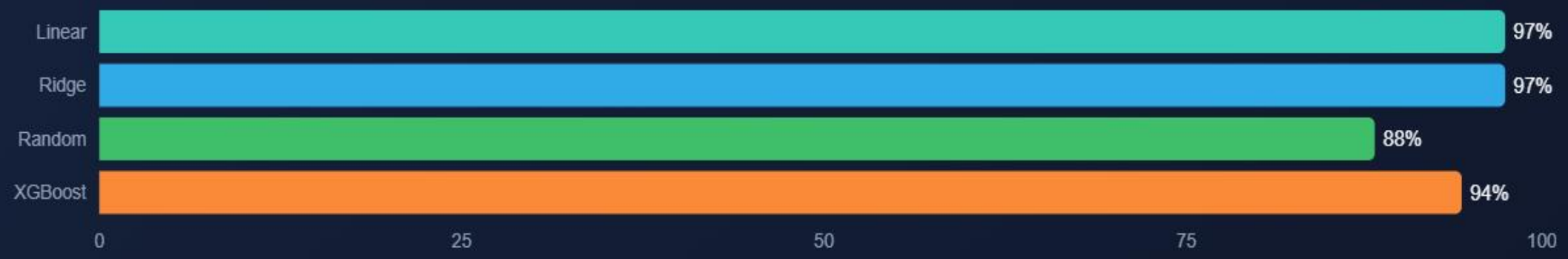
Random Forest

RMSE	≈1.14
R^2	88%

XGBoost

RMSE	≈0.8
R^2	94%

R^2 Score Comparison



Model Comparison

Key insights from comparing algorithm performance

- Linear and Ridge Regression achieved the best performance
- They had the lowest prediction error and highest R^2 score
- Complex models did not outperform simpler models
- The dataset shows a strong linear relationship with the target variable

Best Models: Linear & Ridge Regression

RMSE ≈ 0.54 • R^2 ≈ 97%

This highlights the importance of choosing the right algorithm — simpler isn't always worse!

Model Deployment

From trained model to real-world application

Model Saved
Trained model saved for reuse

→

User Input
Users can input student data

→

Prediction
System returns predicted performance

Practical Application

Deployment transforms the project from a mere model into a usable solution. This step demonstrates real-world applicability and enables both students and educators to benefit from the predictions.

Conclusion & Future Work

Conclusions

- ✓ Machine learning can accurately predict student performance
- ✓ Data preprocessing has a major impact on results
- ✓ Early prediction helps both students and educators

Future Work

- → Use larger and more diverse datasets
- → Add attendance and behavioral features
- → Improve deployment with a user-friendly interface

Thank You!

Questions & Discussion